

Q1. Cloud computing is the delivery of computing services such as servers, storage, databases, networking, and software over the internet (cloud). Instead of owning physical hardware, users can access resources on demand. Key characteristics include on-demand self-service, broad network access, resource pooling, rapid elasticity, and measured service. Popular cloud providers include Amazon Web Services (AWS), Microsoft Azure, and Google Cloud Platform (GCP).

Q2. Public cloud is shared infrastructure available to everyone and is cost-effective but less secure. Private cloud is dedicated to a single organization, offering better security but higher cost. Hybrid cloud combines both, providing flexibility and scalability but adds management complexity.

Q3. Cloud architecture consists of front-end (user interface), back-end (servers, databases), and cloud storage. Virtualization allows multiple virtual machines on one server. Networking connects users and resources securely. Storage manages large data efficiently.

Q4. Security challenges include data breaches, data loss, and insecure APIs. Threats involve unauthorized access and malware. Security measures include encryption, strong access control, multi-factor authentication, and regular security audits.

Q5. Serverless computing allows developers to run code without managing servers. Unlike traditional models, resources are auto-managed. Use cases include event-driven applications, APIs, and real-time data processing.